

.NET 377

Lab1 – The GZ Network (PT)

Answers Document

Due: Sunday, January 18, 2026 by 11:59 pm
50 points total

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Note: all screenshots for this lab should only include CLI commands and outputs and should not include any other part of PT or your computer desktop. Each screenshot must show the command typed and the complete output from each specified command, including the next prompt. You will lose points for any screenshot showing “—More—” at the bottom (you should hit space bar and get complete output).

All pings and traceroutes below must be successful or you will lose points. If any command outputs show “% **Invalid input detected ...**” you will lose points. This means you were in the wrong mode (enable mode, config mode, etc) when you executed the command and must go into the correct mode (using ‘enable’ or ‘configure terminal’ or ‘exit’ etc) before executing the command.

You can edit this file directly or use a different answer file. You should substitute your Student Number for <SN> in all commands.

Fill in All Screenshots and Answers:

1. On GZ-Client, execute a **ping** to **2.2.2.2** (hit Ctrl-C after 3 pings to end output). Take a screenshot and paste here.

```
C:\>ping 2.2.2.2

Pinging 2.2.2.2 with 32 bytes of data:

Reply from 2.2.2.2: bytes=32 time<1ms TTL=253
Reply from 2.2.2.2: bytes=32 time=1ms TTL=253
Reply from 2.2.2.2: bytes=32 time<1ms TTL=253

Ping statistics for 2.2.2.2:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

Control-C
^C
C:\>|
```

2. On GZ-Client, execute **tracert 2.2.2.2**. Take a screenshot and paste here.

```
C:\>tracert 2.2.2.2

Tracing route to 2.2.2.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    33.1.1.254
  2  1 ms    0 ms    1 ms    82.16.5.30
  3  0 ms    0 ms    0 ms    2.2.2.2

Trace complete.
```

3. On Pub-Router, execute a **ping** to **33.1.1.99**. Take a screenshot and paste here.

```
Pub-Router-SH#ping 33.1.1.99
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 33.1.1.99, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
Pub-Router-SH#
```

4. Do a screen capture of the contents of your **Lab1-Static-Routes.xlsx** file and paste into Answers here. Make sure that it is sized so it will be readable.

Static Route Specifications					
Device Name	Destination Network	Destination Mask	Next Hop IP	Outgoing Interface	Comment
GZ-CE-SH	0.0.0.0	0.0.0.0	82.16.5.30	Gi0/0/0	
Pub-Router-SH	0.0.0.0	0.0.0.0	102.19.0.1	Gi0/0/0	
Internet-SH	33.1.1.0	255.255.255.0	82.16.5.29	Gi0/0/0	
Internet-SH	2.2.2.2	255.255.255.255	102.19.0.2	Gi0/0/1	

5. On Internet, execute **show ip interface brief**. Take a screenshot and paste here.

```
Internet-SH#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  82.16.5.30      YES manual  up          up
GigabitEthernet0/0/1  102.19.0.1      YES manual  up          up
GigabitEthernet0/0/2  unassigned      YES unset   administratively down down
Vlan1          unassigned      YES unset   administratively down down
Internet-SH#
```

6. On GZ-CE, execute **show run | include enable**. Take a screenshot and paste here. This will demonstrate that you have set an enable password.

```
GZ-CE-SH>en
Password:
GZ-CE-SH#sh run | include enable
enable password NET377
GZ-CE-SH#
```

7. On Pub-Router, execute **show run | include username**. Take a screenshot and paste here. This will demonstrate that you have set a username and password.

```
Pub-Router-SH#sh run | include username
username Lab1 password 0 NET377!
```

8. On Pub-Router, execute **telnet 33.1.1.<SN> 80**. You should see a line ending in “Open”. Then type “G” and enter, which will close the connection. Take a screenshot and paste here. This will demonstrate that Pub-Router can connect to HTTP service on GZ-Server.

```
Pub-Router-SH>en
Password:
Pub-Router-SH#telnet 33.1.1.19 80
Trying 33.1.1.19 ...Open
[Connection to 33.1.1.19 closed by foreign host]
Pub-Router-SH#
```

9. Take one screenshot showing outputs of this sequence of commands (a-e) on GZ-Server and paste into Answers:
- On GZ-Server, execute **ssh -l Lab1 2.2.2.2**.
 - Enter password for the Lab1 user account
 - Execute **show ip interface brief** which will show interfaces on Pub-Router.
 - Execute **exit** which will close the SSH session.
 - Do a screen capture which shows the interfaces on GZ-Server. You can do a mouse-over of GZ-Server or capture screen shots of the Configuration pages.

GZ-Server

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ip http server
Invalid Command.

C:\>ssh -l Lab1 2.2.2.2

Password:
% Login invalid

Password:

Pub-Router-SH>sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0/0     102.19.0.2      YES manual  up              up
GigabitEthernet0/0/1     unassigned      YES unset   administratively down down
GigabitEthernet0/0/2     unassigned      YES unset   administratively down down
Loopback0                2.2.2.2         YES manual  up              up
Vlan1                    unassigned      YES unset   administratively down down
Pub-Router-SH>exit

[Connection to 2.2.2.2 closed by foreign host]
C:\>
```

The screenshot shows the 'GZ-Server' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the following settings:

- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 33.1.1.19
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 33.1.1.254
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty field) / (empty field)
 - Link Local Address: FE80::209:7CFF:FE97:8689
 - Default Gateway: (empty field)
 - DNS Server: (empty field)
- 802.1X:**
 - ☐ Use 802.1X Security
 - Authentication: MD5 (dropdown menu)
 - Username: (empty field)
 - Password: (empty field)

**** Now, before answering questions 10-14, do the following:**

- i. On GZ-CE router, create an extended Access Control List (ACL) named **NET377-ACL** that denies all TCP packets with destination port = 80 (for all source and destination IPs) and permits all other packets.
- ii. Apply NET377-ACL to incoming packets on GZ-CE router interface Gi0/0/0.

*****;

10. On GZ-CE router, execute **show access-list**. Take a screenshot and paste here.

```
GZ-CE-SH#
GZ-CE-SH#show access-list
Extended IP access list NET377-ACL
 10 deny tcp any any eq www
 20 permit ip any any
```

11. On GZ-CE router, execute **show run | include NET377-ACL**. Take a screenshot and paste here.

```
GZ-CE-SH#
GZ-CE-SH#show run | include NET377-ACL
ip access-group NET377-ACL in
ip access-list extended NET377-ACL
```

12. On Pub-Router, execute a **tracert** to **33.1.1.<SN>**. This should succeed. Take a screenshot and paste here.

```
Pub-Router-SH>en
Password:
Pub-Router-SH#tracert 33.1.1.19
Type escape sequence to abort.
Tracing the route to 33.1.1.19

 1  102.19.0.1      0 msec    0 msec    0 msec
 2  82.16.5.29      0 msec    0 msec    0 msec
 3  33.1.1.19       1 msec    1 msec    0 msec
Pub-Router-SH#
```

13. On Pub-Router, execute **telnet 33.1.1.<SN> 80**. This should **fail** due to the NET377-ACL on GZ-CE router. Take a screenshot and paste here.

```
Pub-Router-SH#telnet 33.1.1.19 80
Trying 33.1.1.19 ...
% Connection timed out; remote host not responding
Pub-Router-SH#
```

14. On GZ-CE router, execute **show access-list**. Take a screenshot and paste here.

```
GZ-CE-SH#sh access-list
Extended IP access list NET377-ACL
 10 deny tcp any any eq www (24 match(es))
 20 permit ip any any (6 match(es))
```

After you have provided all screenshots and answered all questions, save a copy of your Answers document in WORD or PDF format. Look through your file to make sure that all screenshot output is clearly readable.

You should submit TWO documents on D2L for Lab1:

1. Your completed Answers document in PDF format.
2. The PKT file you created in Step 11 of the Lab1 Instructions.